# Feedback Loops

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

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TEACHER GUIDE · 1 OF 9

## Feedback Loops

> Prepare Feedback Loops: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Implement a **deterministic recommender**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case **exposure** check: this source example appears in ai-2/24, ai-2/33. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

### Retrieval answer guidance

### Recommendation Scores

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a weighted score

Reference example, not the only acceptable wording: Score 8.

### Similarity by Selected Traits

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

Look for: Represent items with numeric features

Reference example, not the only acceptable wording: Energy 4, group work 1.

### Missing Data

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Look for: Detect absent and unusable values

Reference example, not the only acceptable wording: 0 is a recorded allowed duration. The other two are unknown under this schema.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

A **recommender** shows poetry in 2 slots and adventure in 20. Poetry gets 1 click; adventure gets 6. Compute clicks per shown slot and explain why neither rate alone proves preference.

### Concept to explain

Recommendations influence behavior, which then becomes new data.

| Term | Meaning |
| --- | --- |
| feedback loop | A cycle in which outputs affect later inputs and therefore future outputs. |
| exposure | The opportunity to see or encounter an item, which can affect later choices. |
| interaction data | Records of what people were shown and how they acted, such as views, clicks or corrections. |
| narrowing | Reducing the allowed choices, scope or interpretation of a task. |

### Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

### Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; **explanation** 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

Teach Week 24 as a two-session investigation using fictional data. Build a fictional activity **recommender** using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name.

This makes the outcome reproducible rather than dependent on **input** order. Assume unique names and validated numeric fields. Assess the three stated outcomes through the independent task, not by copying the worked result.

### Readiness answer

No. Filter by the hard constraint before ranking.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

### End goal for the pictorial case

By the end, I can explain how recommendation **exposure** can create a **feedback loop**.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

## Explain the mechanism

> Model Feedback Loops using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

**Input → Process → Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. System recommends mostly fantasy.

2. Reader can only click among shown books.

3. Clicks are treated as stronger fantasy preference.

4. Next round shows even more fantasy, hiding other possible interests.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

### Concrete teaching case

> **The teaching case**
>
> A **recommender** shows only adventure books, uses clicks to learn and then shows more adventure.

> **The teaching case**
>
> Explain the **feedback loop**.

### Exact explanation to model

> **Why the result follows**
>
> **Exposure** affects clicks; clicks affect future **exposure**. Record what was shown and offer controlled variety.

### Explain the added worked example

Build a fictional activity **recommender** using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name.

This makes the outcome reproducible rather than dependent on **input** order. Assume unique names and validated numeric fields.

## Explain the added worked example · 1

The function prints nothing; the caller prints A. best starts as None to mean no eligible option found. Python short-circuits or: when best is None, later comparisons in that condition need not run.

After selecting an item, best_score is numeric. An empty list or no eligible item returns None.

### Key terms

| Term | Meaning |
|---|---|
| Recommender | A procedure that suggests options using a stated policy |
| Tie | Two options have the same comparison score |
| Deterministic | The same inputs and rules produce the same result |
| Explanation | Evidence describing how a result was obtained |

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **Predict the result if the list order is reversed, and if the limit is 8.**

| Thinking move | Teacher prompt |
|---|---|
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal **explanation** together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Feedback Loops

A fictional catalogue repeatedly recommends two books using observed clicks.

### The facts we will use

**Supplied facts**

Initially each click count is 0. Equal counts choose earlier ID A over B.

**Supplied facts**

Only the recommended book is shown. In an invented three-round record, A is shown and clicked each round; B is never shown.

AI ACADEMY / AI-2/24

**Exposure and clicks after the invented three rounds**

| | Shown | Clicked |
|---|---|---|
| A | 3 | 3 |
| B | 0 | 0 |

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/24

# Feedback Loops

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Initial tie

Counts A=0,B=0; ID rule shows A.



## 2. Record event

A receives a click, so its count becomes 1; B stays 0.



## 3. Repeat selection

Higher A count causes A to be shown again; after three clicks A=3,B=0.



## 4. Interpret carefully

B's zero may mean no exposure, not dislike.

**CHANGE ONE THING**

On a planned trial, show B once and record no click.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Initial tie | Counts A=0,B=0; ID rule shows A. |
| 2. Record event | A receives a click, so its count becomes 1; B stays 0. |
| 3. Repeat selection | Higher A count causes A to be shown again; after three clicks A=3,B=0. |
| 4. Interpret carefully | B's zero may mean no exposure, not dislike. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### Why the result follows

The selection policy changes which **evidence** is available. A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books.

Keep display counts separate from click counts; investigate alternatives with a teacher-approved **exposure** plan rather than treating zero as negative feedback.

### What this example does not establish

#### Keep the stated limit

A click can reflect curiosity or a mistake; even equal **exposure** does not reveal every reason for selection.

### Change one thing in the pictured case

#### Change one thing

Change: On a planned trial, show B once and record no click.

#### Predicted result

**Prediction**: B gains one **exposure** but still zero clicks.

#### Why the result follows

Reason: The **observation** now supports a nonclick on that display, not a permanent dislike.

### Ask while showing the picture

## Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

# Demonstrate and guide practice

> Demonstrate Feedback Loops, reduce support deliberately and check individual reasoning before discussing answers.

### Board sequence

#### 1. Put the facts on the board

Write the given facts: A **recommender** shows only adventure books, uses clicks to learn and then shows more adventure.

#### 2. State the decision to explain

Write the question to explain: Explain the **feedback loop**.

#### 3. Reveal the worked reasoning

Reveal after **prediction**s: **Exposure** affects clicks; clicks affect future **exposure**. Record what was shown and offer controlled variety.

#### Hold back the independent task

Keep for an independent attempt: Only space books are shown, so nearly every click is on space. Can these clicks prove readers dislike other topics? Explain the loop.

### Classroom dialogue

#### Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

### Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an **explanation**.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### Model these guided questions

Reversing still returns A. Limit 8 leaves no eligible item and returns None. Exit: X meets the 12-minute limit and ties for highest eligible score 9; the alphabetical tie rule selects X.

### Case-specific teaching sequence

| Stage | Teacher action |
| --- | --- |
| Session 1 start to 7 minutes | Retrieve the prerequisite and explain the purpose in the opening section. |
| 7 to 20 minutes | Read the worked example and diagram; ask learners to trace each step aloud. |
| 20 to 33 minutes | Complete W1 and W2, then compare predictions with observed results. |
| 33 to 45 minutes | Collect the guided prediction before running; discuss a mismatch and its cause. |
| Session 2 start to 8 minutes | Retrieve the main rule without showing the previous answer. |
| 8 to 20 minutes | Use W3 to diagnose and repair the stated failure; rerun an earlier case. |
| 20 to 35 minutes | Collect W4 independently, including code or plan, predictions and explanations, before hints. |

| Stage | Teacher action |
| --- | --- |
| 35 to 45 minutes | Discuss W5 and the exit; use the fresh retry when support was needed. |

### Reduce help deliberately

#### Completion practice

Completion task: State recommend([], 10) and the worked items with limit 8.

#### Reasoning to listen for

Teacher check: Both return None; the caller should display no eligible option.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for Implementing a **recommender**. Proposed **explanation**: “A successful run proves the conclusion”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

#### Expected repair

Expected correction: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action.

Display the chosen rule and let a person review whether an option is suitable.

Collect **explanation**s, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group **explanation** while others solve a more demanding variation.

### Pictorial practice question

> What extra field would reveal why comparing raw clicks is misleading?

### Reasoned teacher answer

Display or **exposure** count: A shown three times, B zero. Raw clicks confound opportunities with response.

Changed-case reasoning: B gains one **exposure** but still zero clicks. The **observation** now supports a nonclick on that display, not a permanent dislike.

Collect each learner's **explanation** before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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## Run the weekly evidence lab

> Run the Feedback Loops investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Act out three recommendation rounds. Log shown items, choices, and updated profile; add an exploration rule and compare diversity.

> What did the hardest test teach you that the normal test did not?

### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

### Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

A successful run proves the conclusion: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action.

Display the chosen rule and let a person review whether an option is suitable.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an **explanation**. Fresh retry: B (2,4,8), C (3,2,8), A (5,5,9), limit 8. B/C tie at 8; B wins. Limit 7 returns None.

A convenient shortcut preserves meaning: Updating best on score >= best_score makes the last tied item win, so reordering can change results. Use the explicit name comparison for ties.

Test both orders and a no-match case after the repair.

### Match the support to the first error

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral **explanation**, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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## Assess and explain the answers

> Use the Feedback Loops questions, teacher keys and criteria after the relevant attempts.

> Previously discussed cases are practice; protect the separately selected independent task.

### Worked case answer

#### Teacher key · Worked case answer

**Exposure** affects clicks; clicks affect future **exposure**. Record what was shown and offer controlled variety.

### Independent question

#### Independent question · withhold the key

Only space books are shown, so nearly every click is on space. Can these clicks prove readers dislike other topics? Explain the loop.

### Expected independent answer

#### Teacher key · Expected independent answer

No. **Exposure** affects what can be clicked. Showing space produces space clicks that may cause even more space recommendations; test other relevant options to learn more.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “Clicks reveal preferences independently of what is offered.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

#### Teacher key · Original transfer answer

Poetry is 1/2=50%; adventure is 6/20=30%. Raw clicks favour adventure but **exposure** rates differ. Position, user mix and tiny poetry sample confound interpretation. Run a balanced comparison and record **exposure**s and outcomes.

### Explanations for the additional workbook examples

#### W1 Trace selection

#### Workbook question · W1

Why does A win the worked example even though B is **process**ed first?

#### Teacher reasoning · W1

Both score 8 and qualify; A is alphabetically first under the tie rule.

#### W2 Handle no match

#### Workbook question · W2

State recommend([], 10) and the worked items with limit 8.

### Teacher reasoning · W2

Both return None; the caller should display no eligible option.

### Teacher reasoning · W2

For the empty list, the loop has no iterations, so best stays None. With the worked durations 10 and 9 under a limit of 8, neither item passes the eligibility check, so best again stays None.

The caller must turn this result into a clear no-eligible-option message; it must not treat None as an activity name.

## W3 Repair order dependence

### Workbook question · W3

A tied selection changes after reversing **input** order. Identify the likely missing policy and test.

### Teacher reasoning · W3

An explicit **deterministic** tie rule is missing or incorrect. Apply alphabetical name order and rerun both orders.

## W4 Implement independently

### Workbook question · W4

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict

the winner, reverse the list and retest. Also test limit 9.

### Teacher reasoning · W4

X and Y score 9; Z is excluded. X wins alphabetically in either order. Limit 9 returns None.

**W5 Explain the result**

### Workbook question · W5

Write a one-sentence **explanation** of X winning without calling it objectively best.

### Teacher reasoning · W5

X meets the 12-minute limit and ties for highest eligible score 9; the alphabetical tie rule selects X.

**Assess independent transfer**

### Independent question · withhold the key

Selected case: Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

**Selected case answer**

### Teacher key · Selected case answer

X and Y score 9; Z is excluded. X wins alphabetically in either order. Limit 9 returns None.

| Dimension | Evidence required |
| --- | --- |
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect and extend learning

> Connect Feedback Loops to Transparent Book Recommender, adapt support and plan a later changed-case check.

### Transparent Book Recommender

Reserve some varied suggestions and distinguish ‘not shown’ from ‘not liked.’

Keep the decision, **input**, result, **explanation** and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

### Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

### Bridge to the next week

Next: Week 25: Text as Pieces. Retrieve today's **evidence** and identify what carries forward and what is new.

### Support and fresh reassessment

Trace best and best_score row by row. Extension: return a record with name, score and reason while retaining the same selection policy.

Independent W4 **evidence**: X and Y score 9; Z is excluded. X wins alphabetically in either order. Limit 9 returns None. Record whether the learner needed no help, a hint or **explanation**.

Fresh retry: B (2,4,8), C (3,2,8), A (5,5,9), limit 8. B/C tie at 8; B wins. Limit 7 returns None.

Next week: File **input** and **output**. Reuse this week's explicit contracts and **evidence**.

### Plan the delayed return

Suggested return points in this level: ai-2/25, ai-2/27, ai-2/30. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

Log **recommender exposure**, choice and explicit feedback separately; consider whether its own policy narrows future **evidence**.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Feedback Loops lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Implement a **deterministic recommender**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
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> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: State recommend([], 10) and the worked items with limit 8.

### Supported practice answer

Reasoned practice answer: Both return None; the caller should display no eligible option.

Check every learner before discussion: Use the worked case for Implementing a **recommender**. Proposed **explanation**: “A successful run proves the conclusion”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action.

Display the chosen rule and let a person review whether an option is suitable.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken **explanation** or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: On a planned trial, show B once and record no click. Require a **prediction** and a reason before comparing. Expected result: B gains one **exposure** but still zero clicks.

Reason: The **observation** now supports a nonclick on that display, not a permanent dislike.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's **explanation** and **evidence**.

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## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an **explanation**. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-**explanation** links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess **explanation**, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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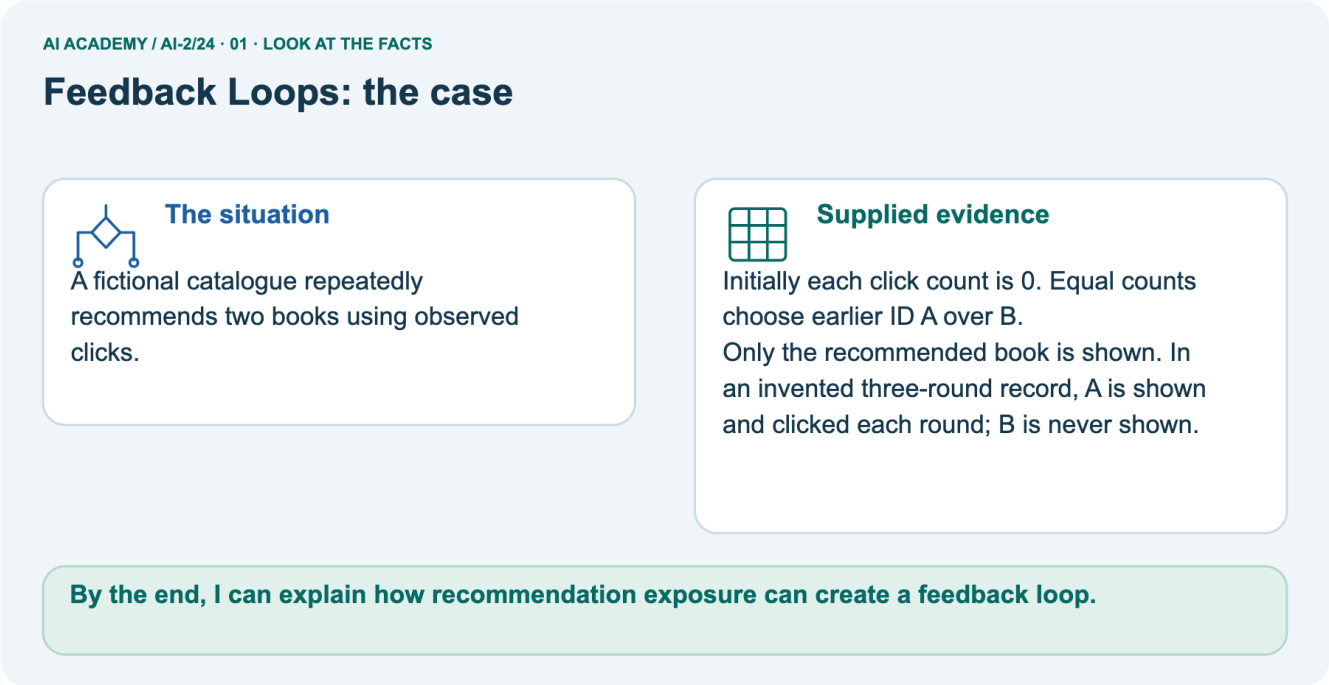


## Visual concept journey

> Show one Feedback Loops graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

## 01. Feedback Loops: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional catalogue repeatedly recommends two books using observed clicks. Supplied facts: Initially each click count is 0. Equal counts choose earlier ID A over B. Only the recommended book is shown.

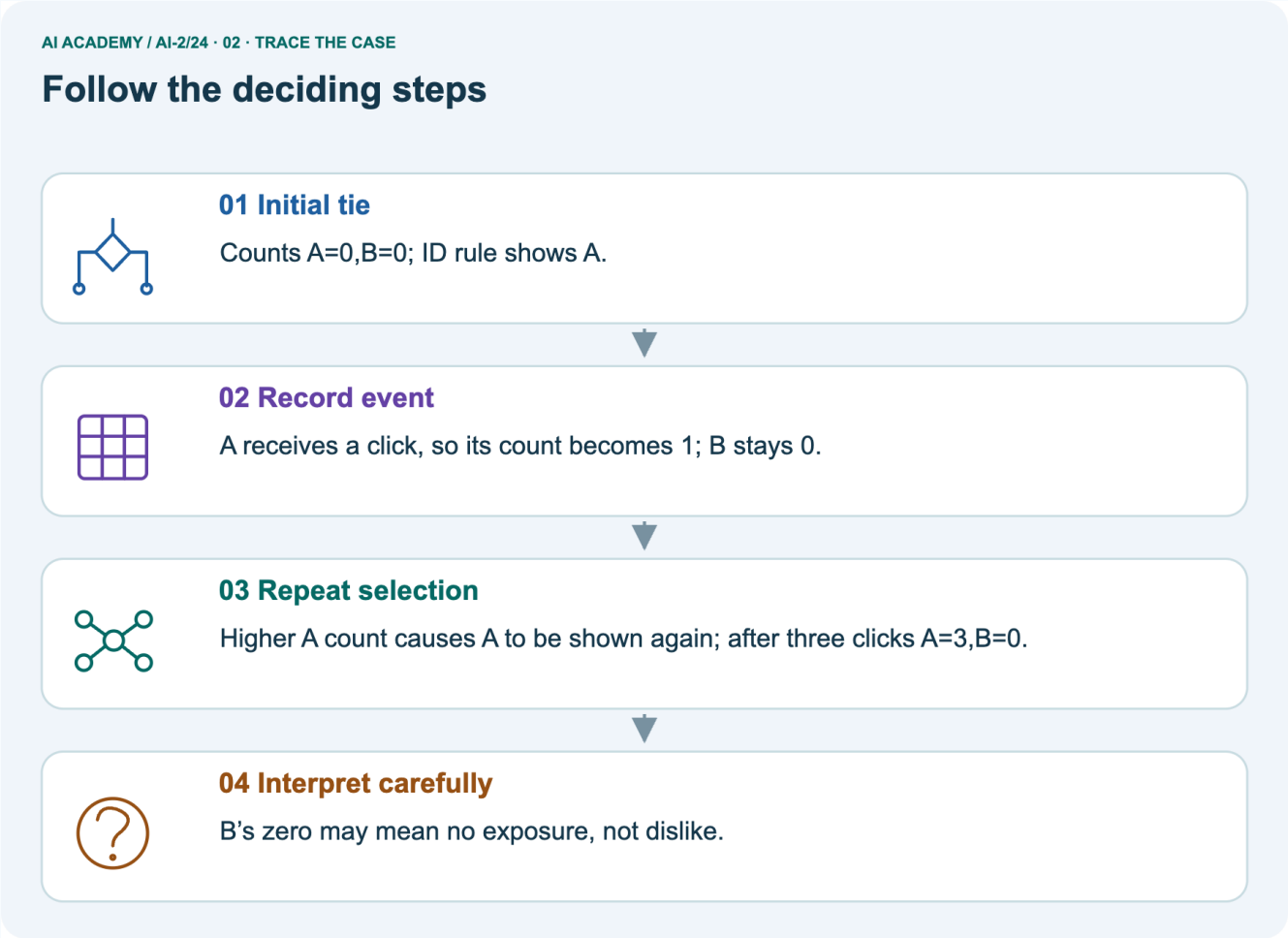
In an invented three-round record, A is shown and clicked each round; B is never shown.

Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

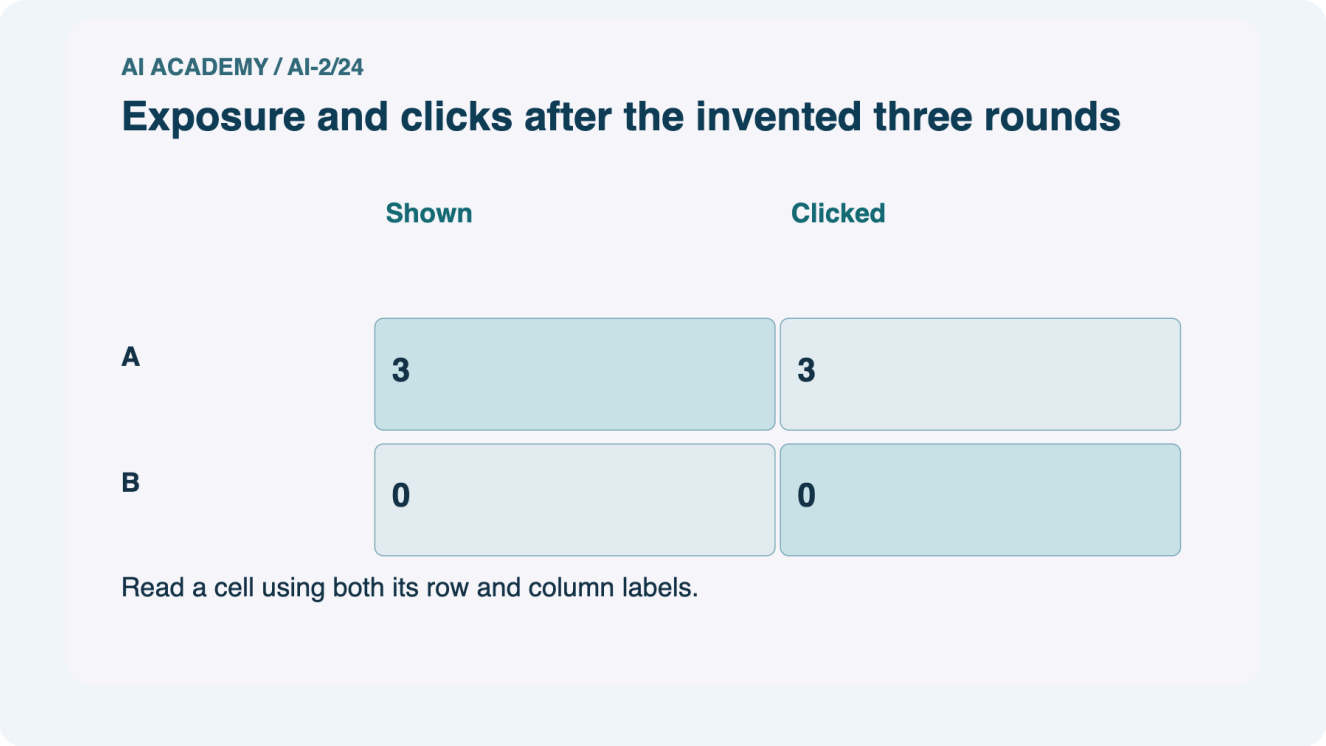
Initial tie: Counts A=0,B=0; ID rule shows A. → Record event: A receives a click, so its count becomes 1; B stays 0. → Repeat selection: Higher A count causes A to be shown again; after three clicks A=3,B=0. → Interpret carefully: B's zero may mean no **exposure**, not dislike.

> Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 03. Exposure and clicks after the invented three rounds




Follow the labelled picture with the matching explanation. Open the vector diagram

**Exposure** and clicks after the invented three rounds. A: Shown = 3, Clicked = 3. B: Shown = 0, Clicked = 0.

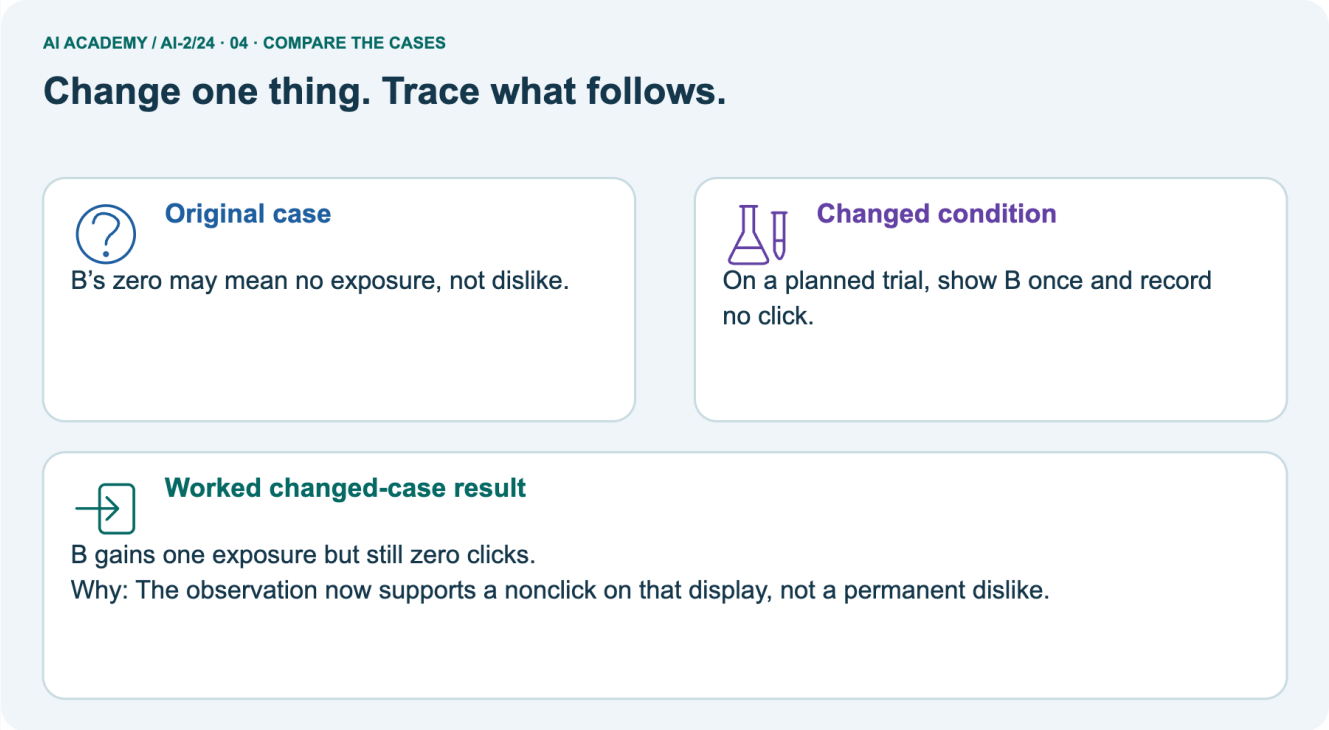
Read the labels and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

**04. Change one thing. Trace what follows.**




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B's zero may mean no **exposure**, not dislike. Changed condition: On a planned trial, show B once and record no click. Expected result: B gains one **exposure** but still zero clicks.

Reason: The **observation** now supports a nonclick on that display, not a permanent dislike.

Predict before checking the worked changed-case result. Name what changed and what the

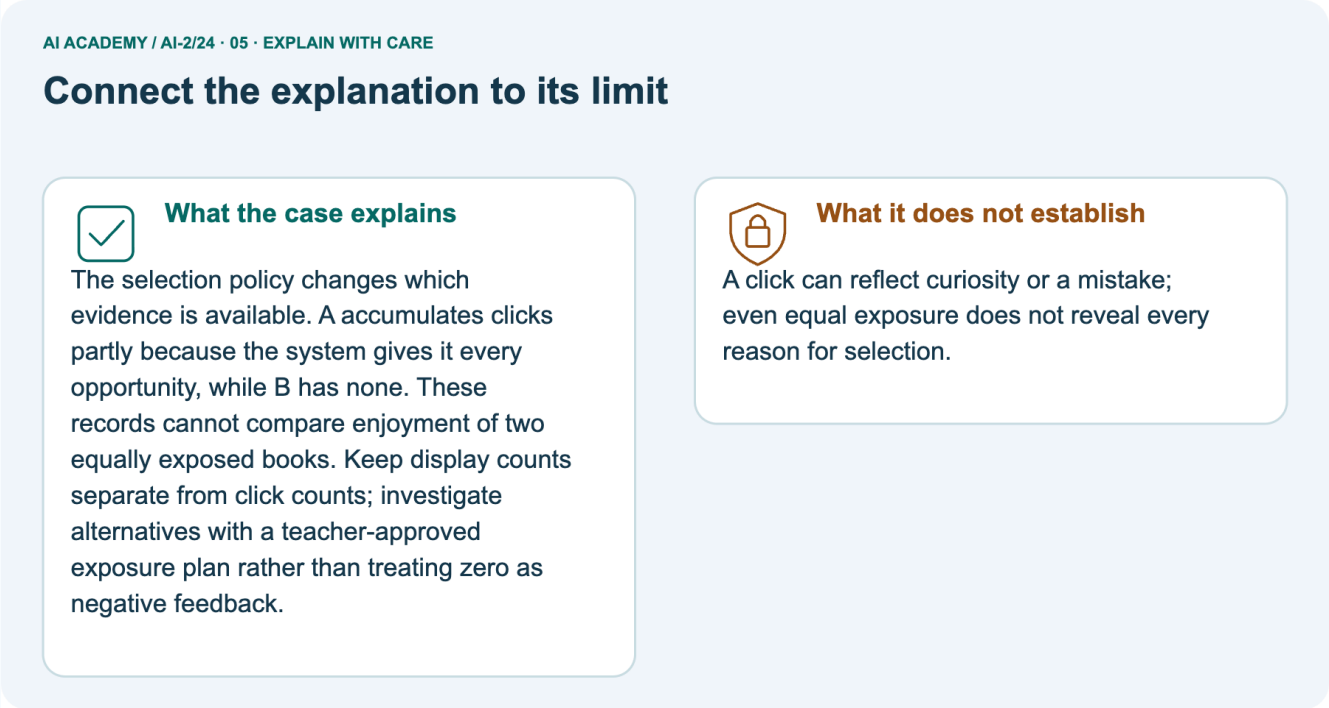
supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The **observation** now supports a nonclick on that display, not a permanent dislike.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

The selection policy changes which **evidence** is available. A accumulates clicks partly because the system gives it every opportunity, while B has none. These records cannot compare enjoyment of two equally exposed books.

Keep display counts separate from click counts; investigate alternatives with a teacher-approved **exposure** plan rather than treating zero as negative feedback.

Limit: A click can reflect curiosity or a mistake; even equal **exposure** does not reveal every reason for selection.

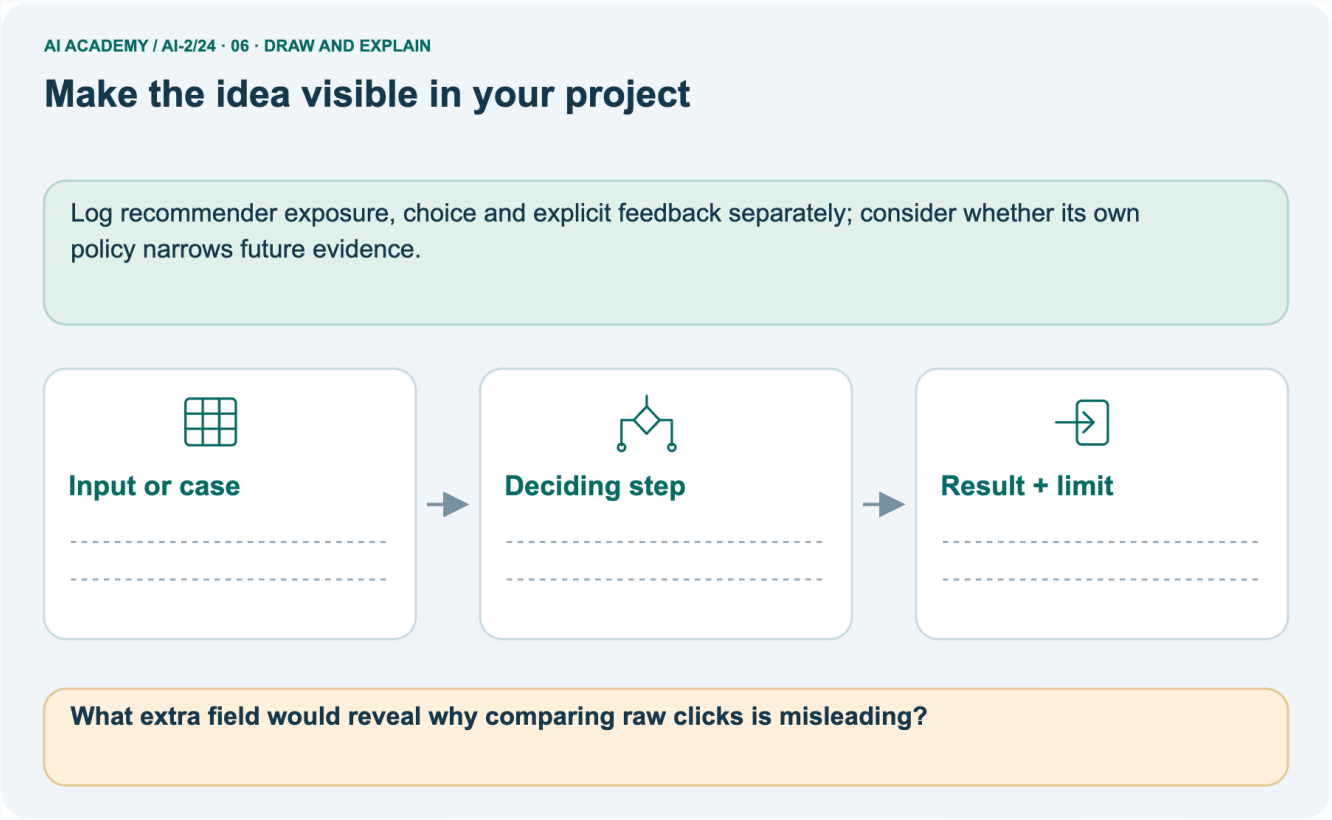
Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A click can reflect curiosity or a mistake; even equal **exposure** does not reveal every reason for selection.

### 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Log **recommender exposure**, choice and explicit feedback separately; consider whether its own policy narrows future **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: What extra field would reveal why comparing raw clicks is misleading?

> Draw your project trace and justify the deciding step using **evidence**.

#### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: Display or **exposure** count: A shown three times, B zero. Raw clicks confound opportunities with response.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.